



Pulsed Nuclear Magnetic Resonance

PHYS3111 Quantum Physics

Operating Instructions

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1 Initial Setup

Before switching anything on:

- Locate the light oil (L-OIL) sample and adjust the rubber o-ring such that the ring's bottom is 43 mm above the bottom of the sample tube.
- Insert the sample tube into the hole in the centre of the permanent magnet (see Figure 1).
- **DO NOT FORCE IT IN UNDER ANY CIRCUMSTANCES** - if it doesn't drop in easily, seek assistance.

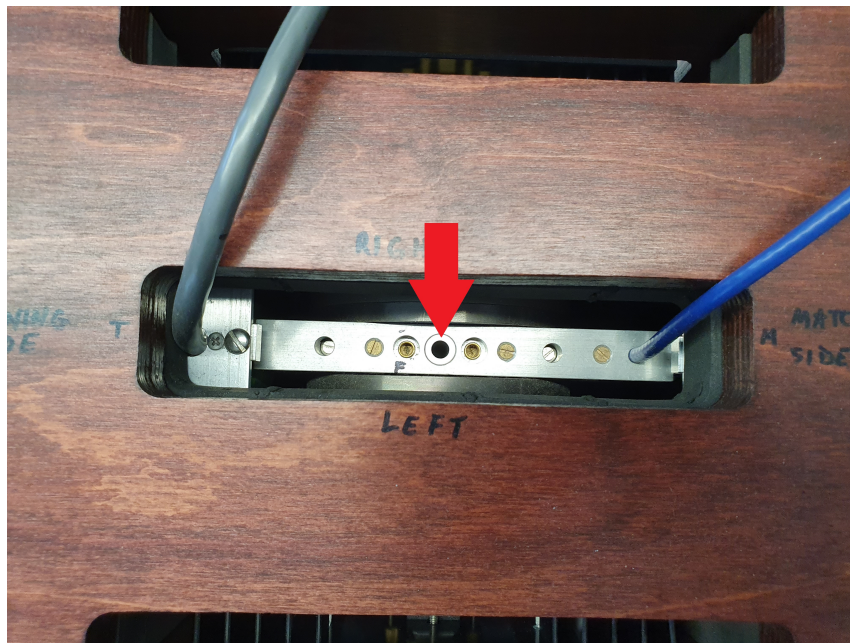


Figure 1: Permanent magnet with location of sample hole indicated.

The following setup is performed with this sample in position, the BNC cables should all be in the correct settings, however their connections are provided for reference.

1.1 PS2 Controller

This controller stabilises the temperature of the permanent magnet and controls the currents through the field gradient coils.

The gradient coil currents should have been left close to optimised by the previous user, but if you're having trouble achieving a free induction decay (in section 2) of ~ 5 ms then you might restore to these default values and adjust from there.

Feedback Loop	Setting
Right	OPEN
Left	OPEN

Gradient Coil	Setting
X	-1.15
Y	-0.80
Z	-0.55
Z ²	-3.55

1.2 21 MHz Receiver

The basic function of the receiver module is to amplify the small voltage induced in the sample coil by the precessing nuclear spin to a large enough amplitude where it can be displayed on an oscilloscope.

Parameter	Setting
Gain	0.75
Band	p
Blanking	On
Width	Position vertically
Filter TC	0.1 ms

1.3 21 MHz Synthesizer

The Synthesizer is the source of the RF signals used in the spectrometer, it has the capability to provide a high stability signal with a wide frequency range. In practice we will only utilise a small frequency range around ~ 21.7 MHz (the approximate resonance frequency of protons and fluorine nuclei in the static magnetic field provided).

Parameter	Setting
CW Out	Off

1.4 Pulse Programmer

The Pulse Programmer determines the duration of individual RF pulses as well as the number of pulses in a series, the spacing between pulses, and how often an entire series is repeated. The PS2 pulse programmer provides two different pulses, A and B, both at the same frequency. Duration and number of pulses, however, are independent. The selected pulse pattern is sent from the Pulse Out connectors of the Pulse Programmer to the Pulse In connectors of the Synthesizer. For the experiments in this discussion, Sync Out connects to the oscilloscope. The Sync toggle switch can be set to either pulse. Now let's look at how each of the pulse parameters is controlled by the settings on the Pulse Programmer.

A_{len} and B_{len} control the length of time each of these RF pulses persists. (In the Magnetic Torque apparatus, this "pulse" time is equivalent to how long you must rotate the horizontal field to get the ball's handle, and thus its magnetic moment, horizontal.)

N indicates the number of pulses to be used. The PS2 provides only one A pulse, but there can be anywhere from 0 to 100 B pulses in a given series.

The τ setting indicates the delay time, the time between the first and second pulses of a series. When more than two pulses are used, the system adjusts subsequent delay times between the second and third, third and fourth etc. to 2τ . We will see why this matters later.

The P , or Period setting determines how often an entire pulse sequence is repeated. This is also called the repetition time.

The total time for the pulse series itself, is determined by the combination of N and τ . This must be less than the repetition time P so that the digital logic does not lock up. In addition, P must be long enough so that, after the pulse series had ended, the net magnetisation has time to realign with the primary field. If P is too short, a pulse sequence will begin before the system has returned to thermal equilibrium and the FID will make little sense. In fact the repetition time P , should be such that the time after the last echo or FID is long compared to T_1 . To be safe it should be close to $10 T_1$.

Parameter	Setting
MG	Off
Sync	A
Pulse A	On
Pulse B	Off

1.5 BNC Connections

Device	Connector		Device	Connector
Pulse Programmer	I Pulse Out	⇔	21 MHz Synthesizer	I Pulse In
Pulse Programmer	Q Pulse Out	⇔	21 MHz Synthesizer	Q Pulse In
Pulse Programmer	Sync Out	⇔	DSO	Ext. Trig.
Pulse Programmer	Blanking Out	⇔	21 MHz Receiver	Blanking In
21 MHz Synthesizer	Pulsed RF Out	⇔	21 MHz Receiver	Pulsed RF In
21 MHz Synthesizer	Ref Out	⇔	21 MHz Receiver	Ref In
Permanent Magnet		⇔	21 MHz Receiver	Sample
21 MHz Receiver	I Out	⇔	DSO	Ch. 2
21 MHz Receiver	Env. Out	⇔	DSO	Ch. 1

2 Free Induction Decay

Once initial setup has been completed, you're ready to observe an NMR signal from the light oil sample, turn on the PS2 spectrometer (switch on rear left) and oscilloscope.

Adjusting some parameters on the Synthesizer and Pulse Programmer requires the use of an adjustment knob, 'Push to Select', and accompanying display.

To adjust the RF frequency (F) and the delay time τ (tau) perform the following steps on the 'Push to Select' knob:

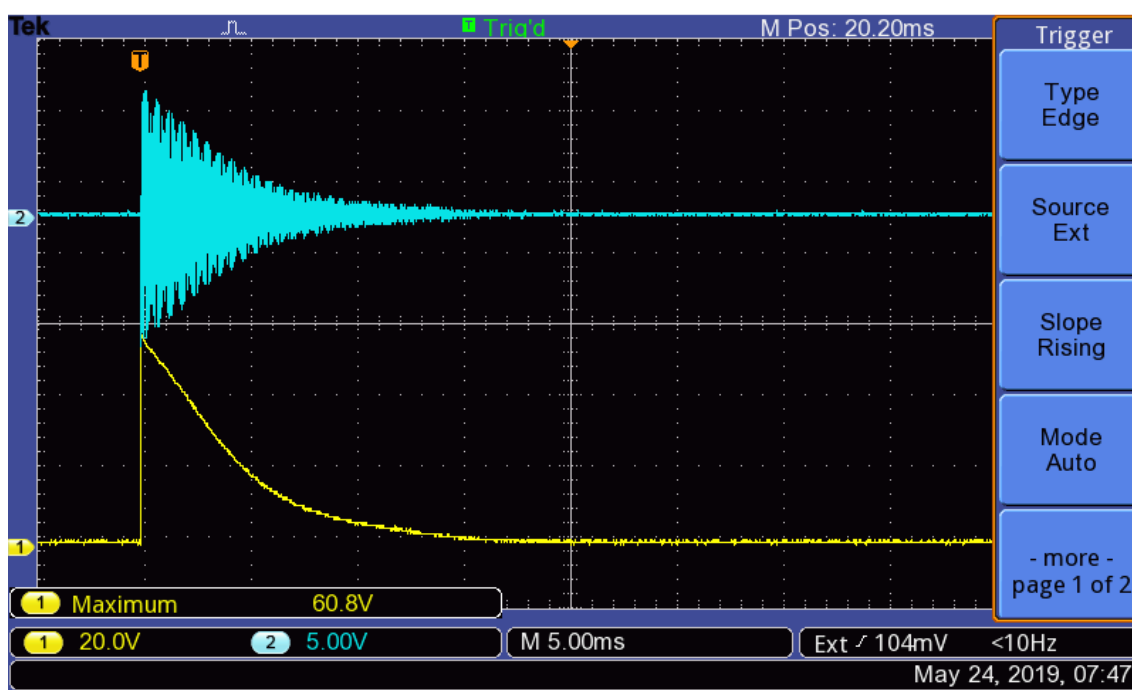
1. Rotate the knob to select (flashing cursor on top line of display) which parameter to adjust and push to select, there will be a beep and the cursor will move to bottom line of display.
2. Now rotate the knob to select which decimal place to be adjusted and press to select (accompanied by beep).
3. Rotate the knob to select value and press to lock in (accompanied by beep).
4. Continue adjusting decimal places (i.e. repeat steps 2-3) or return to the parameter selection (top line) by selecting the up arrow.

To adjust all other parameters on the Pulse Programmer performs steps 1 and 3 above.

Set the following parameters:

Device	Parameter	Symbol	Value
21 MHz Synthesizer	Frequency	F	21.70000 MHz
Pulse Programmer	Length of Pulse 'A'	A_len	2.5 μ s
Pulse Programmer	Length of Pulse 'B'	B_len	0.02 μ s
Pulse Programmer	Number of 'B' Pulses	Num_B	0
Pulse Programmer	Period of sequence	Period	300ms

On the oscilloscope screen you channel 1 should show free induction decay similar to that seen in figure 2.



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Figure 2: Free Induction Decay (FID) before magnetic field gradient adjustment and synthesizer frequency adjustment

2.1 Temperature Adjustment

On the PS2 controller adjust the Left and Right Feedback Loop potentiometers such that the Error Signal Indicators (LEDs) are illuminated (red means the potentiometer is set too high, green means too low). When the LEDs are both unilluminated, close the feedback loops.

The temperature feedback loops should not need adjustment throughout the experiment, however if you're having trouble maintaining the Larmor frequency later in the experiment, it may be worth opening the feedback loops and repeating this adjustment.

2.2 Frequency Adjustment

On the 21 MHz Synthesizer adjust the synthesizer frequency (F) using the 'Push to Select' knob, this will change the signal displayed on Ch. 2. The trace shows an oscillation decaying over a similar time to the FID on Ch. 1, the frequency of the oscillation is the beat frequency (frequency difference) between the synthesizer frequency and the Larmor frequency.

In order to set the synthesizer frequency to the Larmor frequency, the following steps are usually followed:

- Begin by adjusting the 0.001 MHz digit to give the maximum amplitude for the Ch. 2 trace.
- Then adjust the 0.00001 MHz digit carefully to increase the period of the oscillation.
- As you approach the Larmor frequency you should be able to remove the oscillation completely, leaving only a decay trace, similar in shape to that found on Ch. 1 as seen in figure 3.

This frequency adjustment will need to be checked and re-adjusted throughout the experiment, though it should be reasonably stable once the electronics and magnet have had time to warm up.

2.3 Magnetic Field Gradient Adjustment

The magnetic field in this experiment is supplied by a permanent magnet. To make the magnetic field over the sample's volume as uniform (homogeneous) as possible, four small coils, which can generate magnetic field gradients, are situated near the sample position. The field gradients are adjusted by varying the current through the gradient coils.

Adjust the X, Y, Z and Z^2 potentiometers to lengthen the FID tail, until you achieve the longest FID tail possible (while maintaining a roughly exponential decay profile), at least 4 ms for the T_2^* time. See figure 3 as an example.

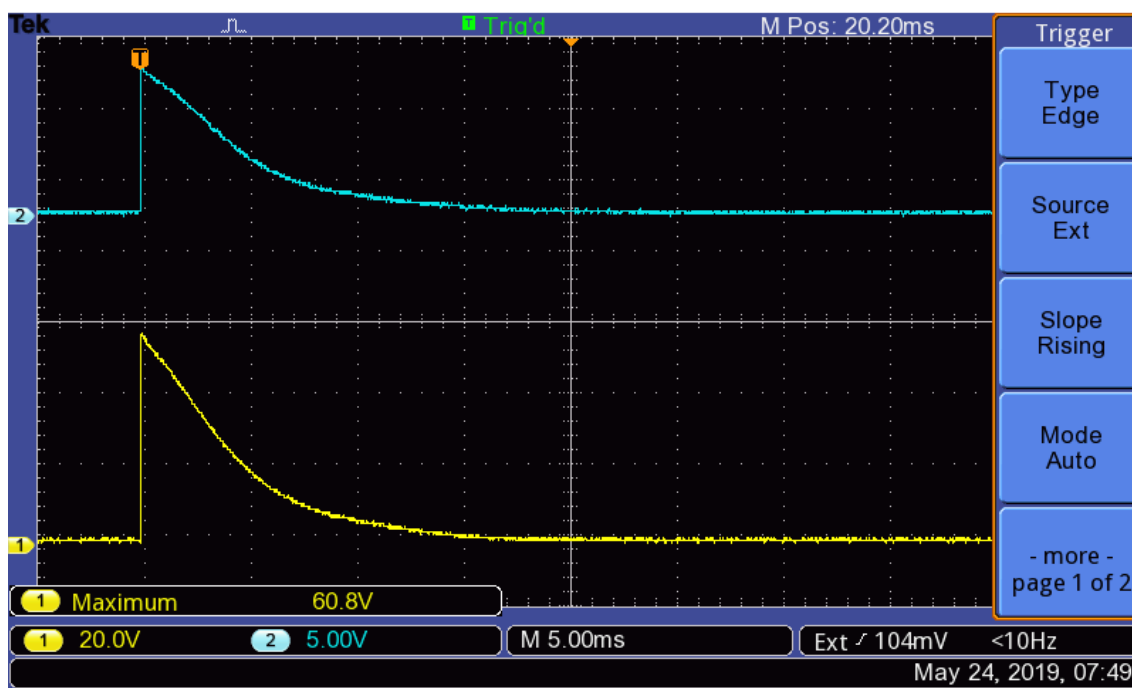
2.4 Final Check

As a final check of your setup, increase the value of A_{len} until you see a minimum in the FID amplitude (around 5 μ s), if the minimum amplitude is above 5 V consult a demonstrator/lab staff as they may need to perform a magnetic yoke adjustment (as detailed in the manufacturer's notes).

DO NOT ATTEMPT TO PERFORM THIS ADJUSTMENT YOURSELF

Further details on the Spectrometer may be found in the Instructor's Manual [1]

3 Oscilloscope



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Figure 3: Free Induction Decay (FID) after magnetic field gradient adjustment and synthesizer frequency adjustment

3.1 Trigger

As mentioned above the above the Pulse Programmer - Sync Out should be connected to the external trigger. The trigger settings should be as follows:

Setting	Value
Trigger Level	~ 100 mV
Type	Edge
Source	External
Slope	Rising
Mode	Auto
Filtering	HF Reject

4 LabVIEW Program

A custom LabVIEW program has been written for this experiment in order to interface with the oscilloscope and make data acquisition and processing a bit easier for you.

The program, titled 'pNMR-v1e.vi' should be located on the desktop and should open automatically in LabVIEW 2018 64-Bit.

Upon opening the program, the first step is to run the program by pressing the white arrow

in the top left of the window.

The program is split into tabs for different parts of the experiment. The operation and explanation of these parts is below. Each of the tabs reads the oscilloscope waveform on the selected channel, and then performs fitting, peak detection or other basic operations. Channel 1 should be selected for all operations other than measuring the Chemical Shift, for which you will use Channel 2.

4.1 T₂*

This tab is used to acquire the Free Induction Decay signal of the current sample and fit an exponential decay to that signal in order to extract a value for T₂* (the FID time constant).

With **Channel 1** selected, press **Collect** to acquire the current waveform. For best results the decay should take up most of the oscilloscope screen (i.e. choose appropriate time-base) and the start of the decay should be near, but not exactly at, the left hand side of the oscilloscope screen (i.e. choose appropriate horizontal offset).

The program will find the maximum value and assume this is the start of the decay (it will place the cursor 'Fit Start Point' at this maximum). Press **Fit** to fit an exponential decay, of the form:

$$y = a \times \exp\left(-\frac{t}{T_2^*}\right) + b \quad (1)$$

to the waveform. The cursor 'Fit Start Point', may be moved to other points along the waveform to start the decay fit there.

Press **Save** to save the raw data as well as the fitting parameters in a text file, and also the 'T₂* Graph' as a bmp image.

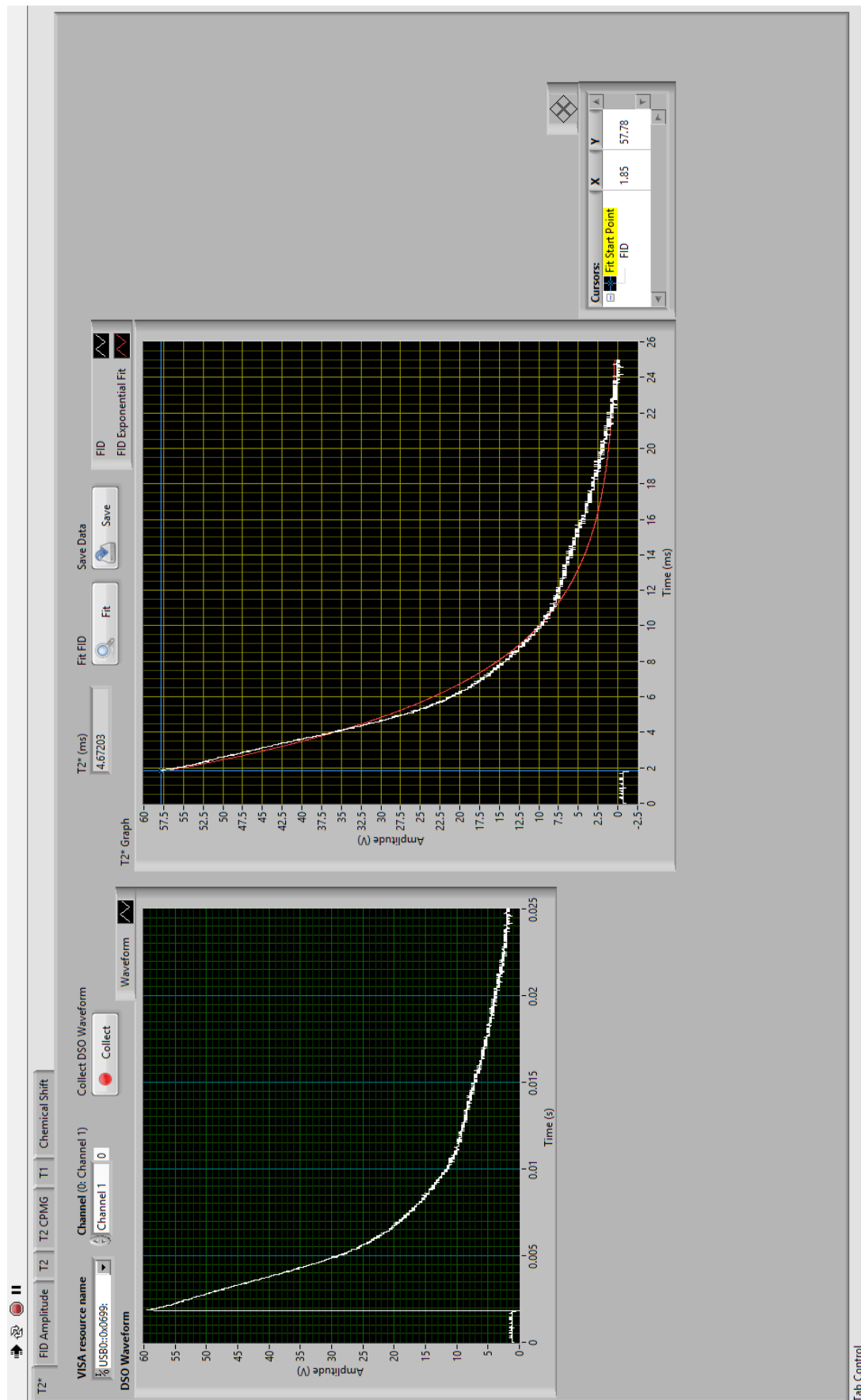


Figure 4: Screen grab of the T2* Tab showing an appropriate Free Induction Decay

4.2 FID Amplitude

This tab is used to acquire the amplitude of the FID as a function of the RF pulse length.

There is usually data already in the 'FID Data' Table as an example, so begin by clearing the table by pressing **Clear**. Now enter the desired length of the RF pulse into the field 'A_len (us)' and set the same value on the Pulse Programmer.

On the oscilloscope you should see a FID curve on Channel 1 (at most pulse lengths), this decay should be replicated on Channel 2 (see Figure 3). For certain pulse lengths $\gtrsim 5\mu\text{s}$ the decay on Channel 2 will be inverted, when this occurs, toggle the **Amplitude Sign** button to 'Negative'.

With **Channel 1** selected, press **Collect** to acquire the current waveform.

The program will find the maximum value of the decay curve (subtracting the background) and adds it to the table with either a positive or negative sign based on **Amplitude Sign**. All values in the table should be automatically plotted.

If you make a mistake you can remove the last row of the table by pressing **Remove**.

Set a range of values in the field 'A_len (us)' (and on the Pulse Programmer) and collect the FID amplitude to complete the table/graph.

Press **Save** to save the tabulated data as well as the 'FID Amplitude Graph' as a bmp image.

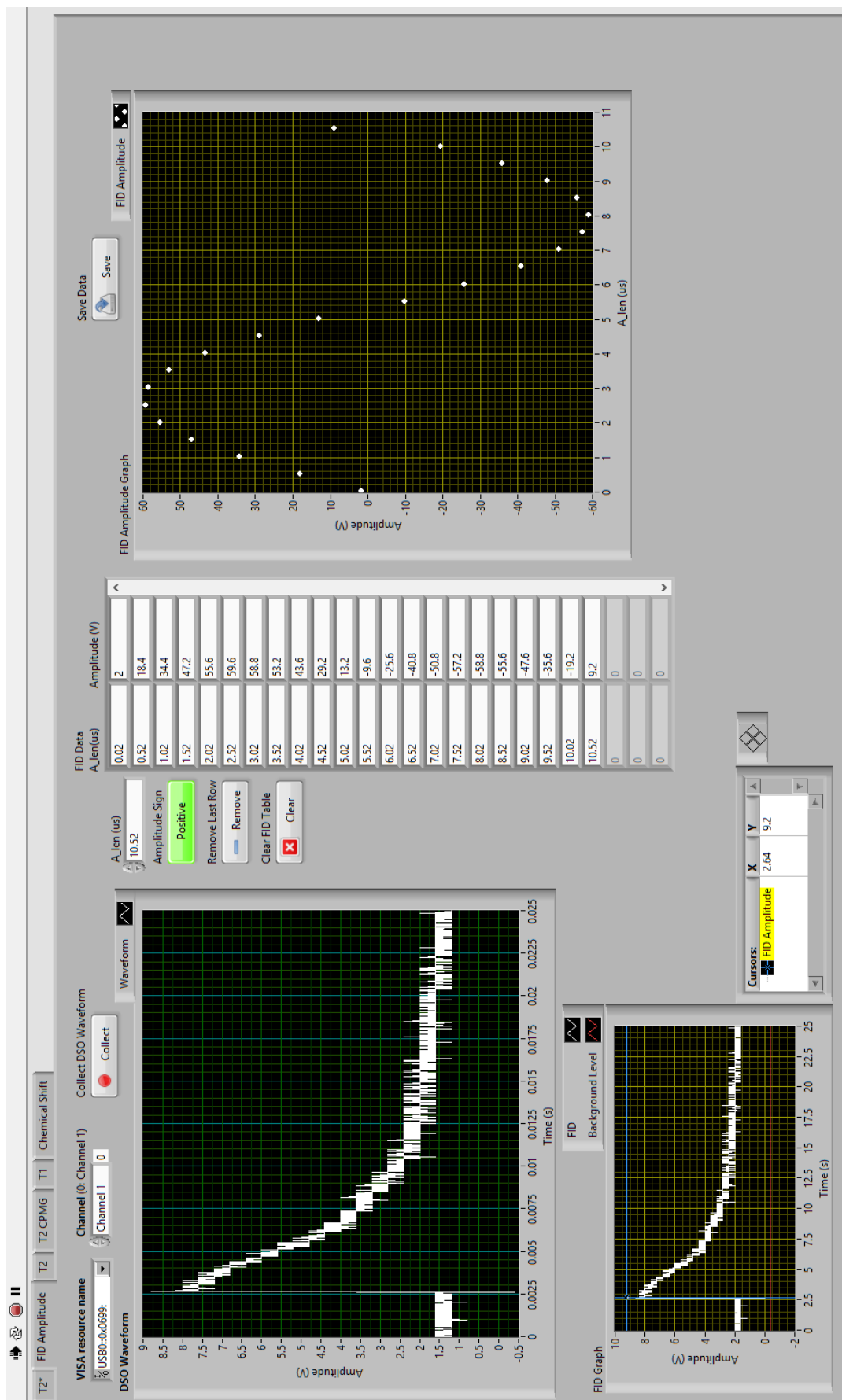


Figure 5:

4.3 T₂

This tab is used to acquire the amplitude of the spin echo as a function of the time (τ) between the 90° pulse (A) and the 180° pulse (B) and then extract the value of the spin-spin relaxation time (T_2).

There is usually data already in the 'T2 Data' Table as an example, so begin by clearing the table by pressing **Clear**. Now enter the desired length of the τ into the field 'Current Tau (ms)' and set the same value on the Pulse Programmer.

With **Channel 1** selected, press **Collect** to acquire the current waveform. The software uses the value of 'Current Tau (ms)' to determine the region (shown on the 'Spin Echo Graph') to search for the spin echo (located 2τ ms after the 90° pulse) and takes the maximum value (indicated on the cursor) and adds it to the table

Set a range of values in the field 'Current Tau (ms)' (and on the Pulse Programmer) and collect the spin echo amplitude to complete the table/graph.

If you make a mistake you can remove the last row of the table by pressing **Remove**.

The data in the 'T2 Data' table is automatically plotted and can be fit with the function:

$$y = a \times \exp\left(-\frac{2\tau}{T_2}\right) + b \quad (2)$$

by pressing **Fit**, to determine T_2 .

Press **Save** to save the raw data as well as the fitting parameters in a text file, and also the 'T2 Graph' as a bmp image.

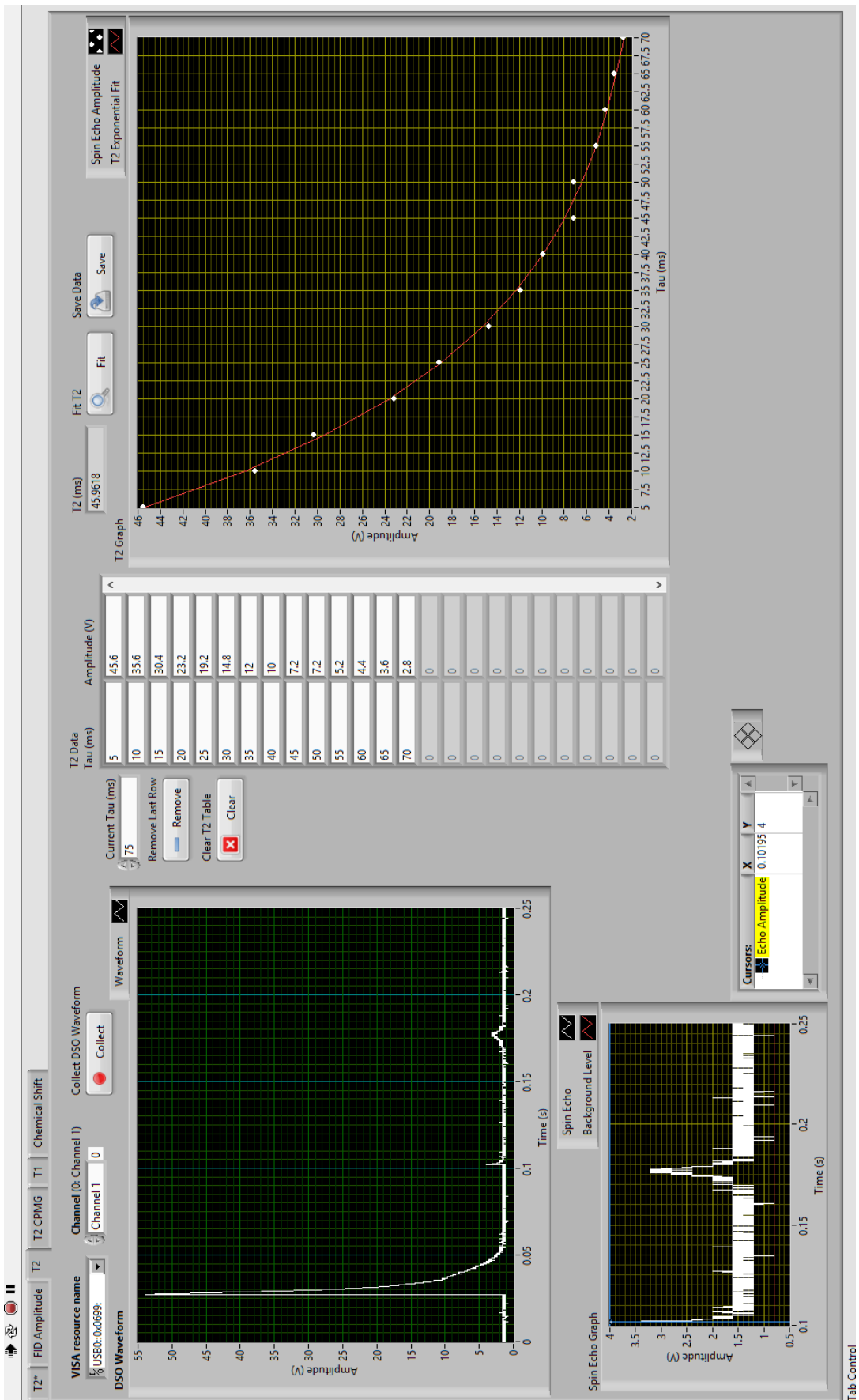


Figure 6:

4.4 T2 CPMG

This tab is used to acquire the amplitude of a sequence of spin echoes and then extract the value of the spin-spin relaxation time (T_2) from a single measurement.

There is usually data already in the 'T2 CPMG Data' Table as an example, so begin by clearing the table by pressing **Clear**. Now enter the desired length of the τ into the field 'Tau (ms)' and the desired 'Number of B Pulses' then set the same values on the Pulse Programmer.

With **Channel 1** selected, press **Collect** to acquire the current waveform, which will then be plotted in 'Spin Echo Graph CPMG'. Press **Find** to search the waveform for the peak positions and their amplitudes. The program searches for a maximum at positions $2n\tau$ after the first peak (where n ranges from 1 to 'Number of B Pulses'), the value of 'Search Width' may need to be adjusted if peaks are erroneously detected, else you can manually edit the values in 'T2 CPMG Data'.

The data in the 'T2 CPMG Data' table is automatically plotted and fit with the function:

$$y = a \times \exp\left(-\frac{\tau}{T_2}\right) + b \quad (3)$$

to determine T_2 .

Press **Save** to save the raw data as well as the fitting parameters in a text file, and also the 'T2 Graph CPMG' and 'Spin Echo Graph CPMG' as bmp images.

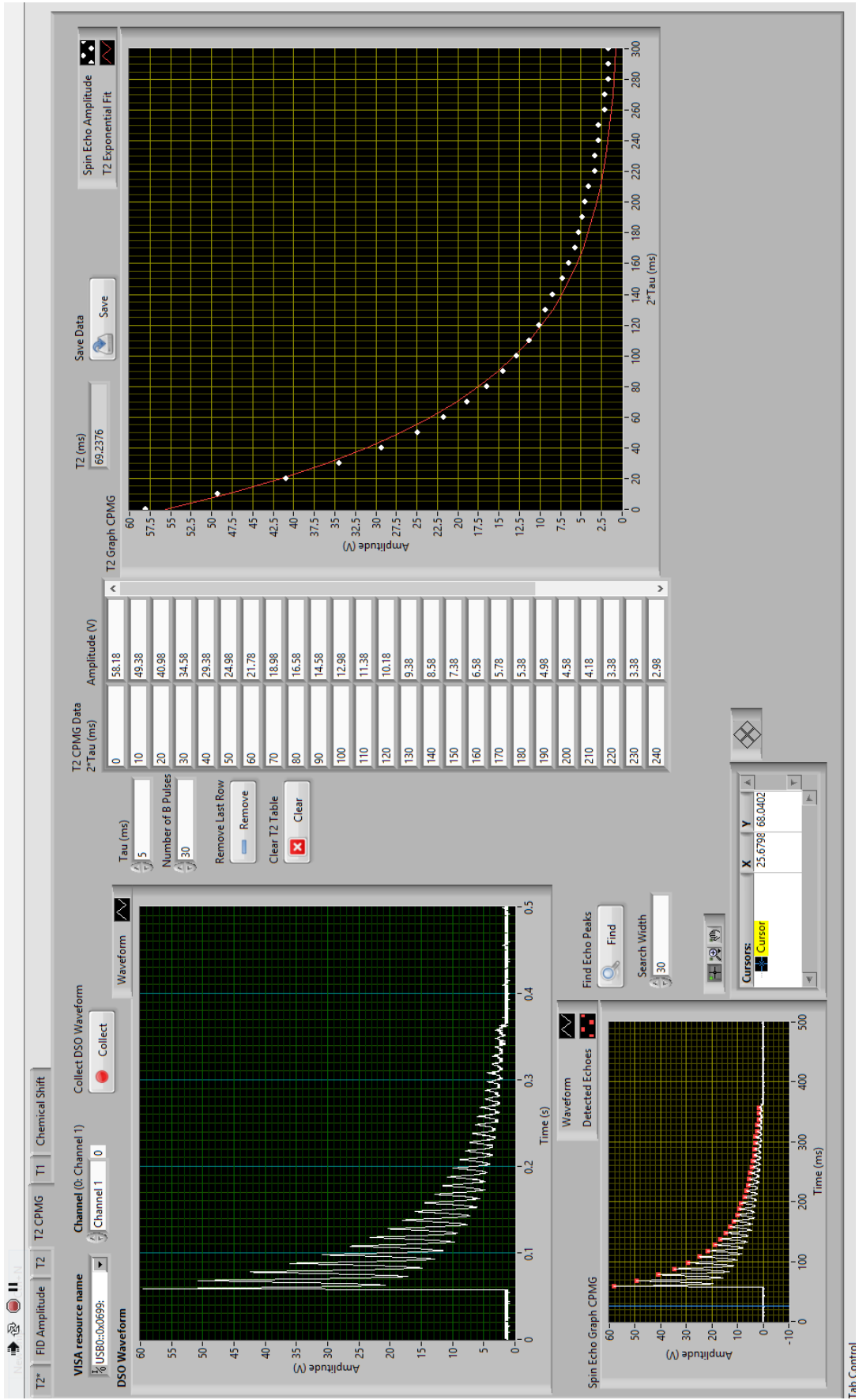


Figure 7:

4.5 T1

This tab is used to acquire the amplitude of the inversion recovery as a function of the time between the 180 pulse (A) and the 90 pulse (B) and then extract the value of the spin-lattice relaxation time (T1).

There is usually data already in the 'T1 Data' Table as an example, so begin by clearing the table by pressing Clear. Now enter the desired length of the into the field 'Current Tau (ms)' and set the same value on the Pulse Programmer.

With Channel 1 selected, press Collect to acquire the current waveform. We're syncing to the B pulse, so the FID will not move in time as increases and the software simply takes the maximum value (indicated on the cursor) and adds it to the table. Set a range of values in the field 'Current Tau (ms)' (and on the Pulse Programmer) and

collect the FID amplitude to complete the table/graph.

If you make a mistake you can remove the last row of the table by pressing Remove.

The data in the 'T1 Data' table is automatically plotted and can be fit with the function:

$$y=a*(1-2*\exp(-x/b))+c$$

by pressing Fit, to determine T1.

Press Save to save the raw data as well as the fitting parameters in a text file, and also the 'T1 Graph' as a bmp image.

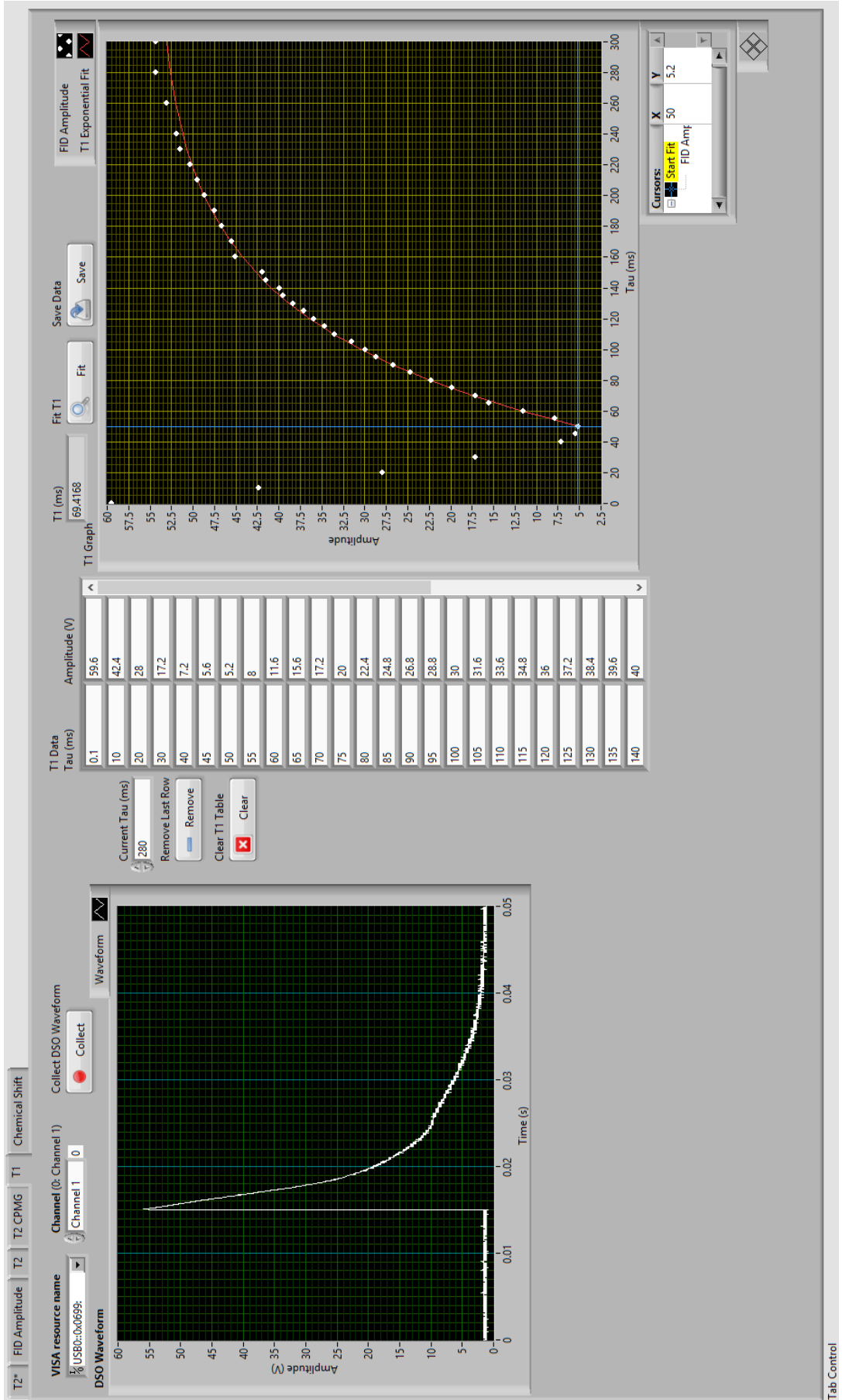


Figure 8:

4.6 Chemical Shift

This tab is used to take the Fourier Transform of the waveform on channel 2 of the oscilloscope when the Synthesizer is purposefully tuned away from the Larmor frequency. This allows is to see resonance peaks from nuclei in in-equivalent sites on the molecule.

Select how many averages to perform using 'Number of Averages' this sets the number of waveforms which are acquired and averaged before taking the fast Fourier transform (FFT). Now with **Channel 2** selected, press **Collect** to acquire the current waveform, wait for acquisition to finish (may take a minute or so depending on number of averages).

The software automatically takes the FFT of the averaged waveform, using the zoom tool (Magnifying Glass), select the region of the FFT where the resonance peaks occur.

Press **Save** to save the raw FFT data as well as the 'Fourier Transform' graph as a bmp image.

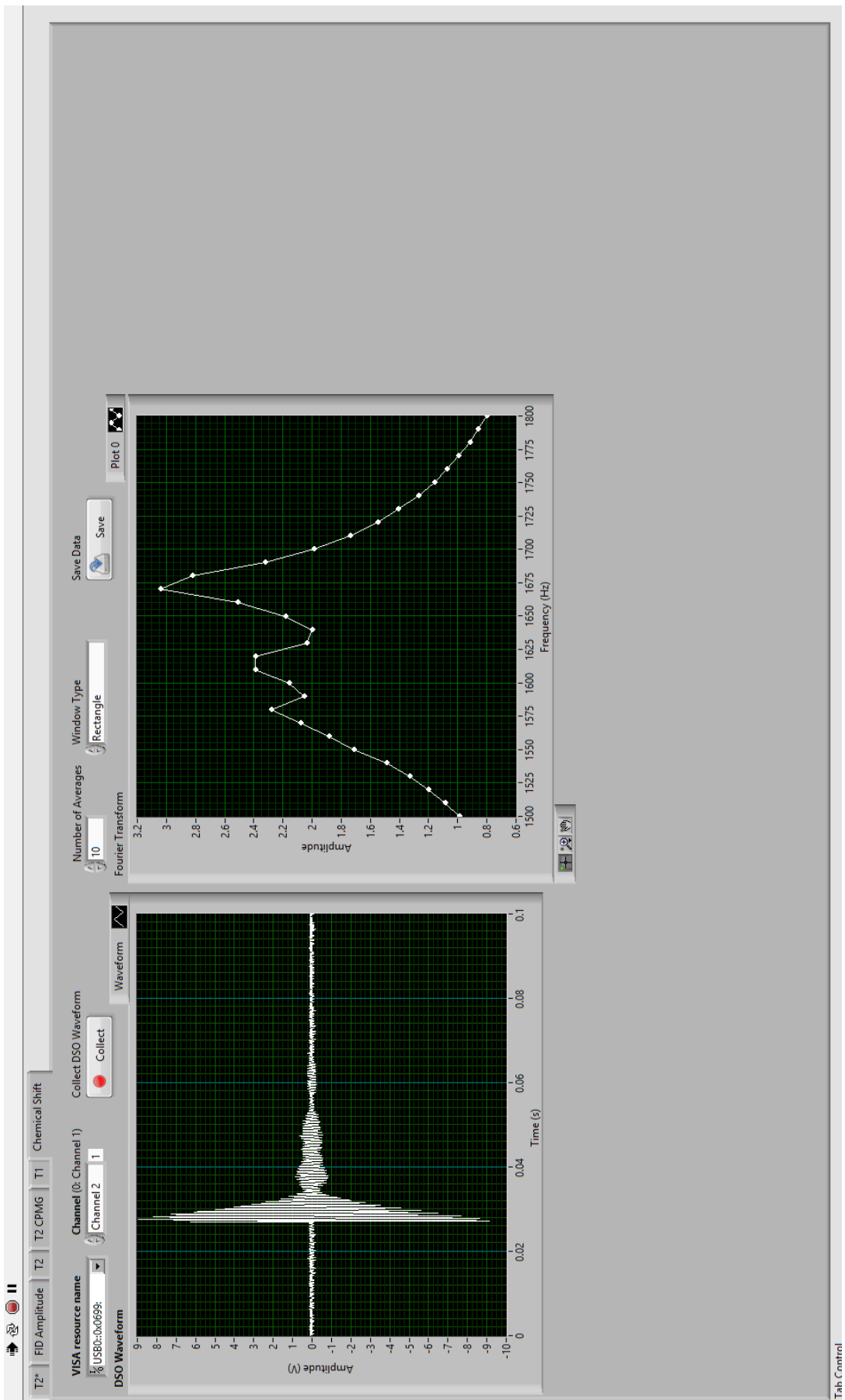


Figure 9:

References

- [1] *Pulsed/CW NMR Spectrometer (PS2-B)*. Instructor's Manual. TeachSpin, Inc.